

Importance of algorithms and Specifications

Module I

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What is an algorithm?

- The word algorithm comes from the name of a Persian author, Abu Jafar Mohammad Ibn e Musa Al Khowarizmi (825 AD), who wrote a textbook on mathematics.
- **Definition:** An algorithm is a finite set of instructions that accomplishes a particular task.
- **Criteria:**
 1. **Input:** Zero or more quantities are externally supplied.
 2. **Output:** At least one quantity is produced.
 3. **Definiteness:** Each instruction is clear and unambiguous.
 4. **Finiteness:** If we trace out the instructions of an algorithm, then for all cases, the algorithm terminates after a finite number of steps.
 5. **Effectiveness:** Every instruction must be very basic so that it can be carried out using pen and pencil. It also must be feasible.

What is an algorithm? Cntd..

- Algorithms that are definite and effective are also called computational procedures. E.g. OS.
- To achieve the criteria of definiteness, algorithms are written using programming languages.
- A program is an expression of algorithm in a programming language.
- The study of algorithms includes many important and active areas of research. Four distinct areas are,
 - 1) How to devise algorithms? [Designing a new algorithm using different design techniques (E.g. Divide & Conquer, Greedy, Backtracking etc.)]
 - 2) How to validate algorithms? [Correctness of algorithm]
 - 3) How to analyse the algorithms? [Computing time & Storage requirements]
 - 4) How to test a program? [Using debugging and profiling]

Algorithm Specification

- Pseudocode conventions:

1) Comments begin with // and continue till the end of line.

2) Blocks are indicated with matching braces. { and }. A compound statement can be represented as a block. The body of a procedure also forms a block. Statements are terminated by (delimited) semicolon.

3) An identifier begins with a letter. The data types of a variables are not explicitly declared. The type of a variable and its scope (local or global) is evident from the context. Compound data types can be formed with records,

```
node = record
```

```
{
```

```
    datatype_1 data_1;
```

```
    ....
```

```
    datatype_n data_n;
```

```
    node *link; // here link is a pointer to the record type node.
```

```
} //Individual items of a record can be accessed with → and period.
```

Algorithm Specification cntd..

- Pseudocode conventions:

4) Assignment of values to variables is done using the assignment statement.

`<variable> := <expression>;`

5) There are two Boolean values true and false. In order to produce these values, the logical operators **and**, **or** and **not** and the relational operators $<$, $\leq$, $>$, $\geq$, $=$, $\neq$ are provided.

6) Elements of multidimensional arrays are accessed using [and]. If A is a two dimensional array, the $\{i,j\}$ th element of array is denoted as $A[i,j]$. Array indices starts at zero.

Algorithm Specification cntd..

- Pseudocode conventions:

7) The following looping statements employed.

```
repeat
    <statement 1>
    .....
    <statement n>
until <condition>
```

```
while <condition> do
{
    <statement 1>
    .....
    <statement n>
}
```

```
for variable := value1 to value2 step step do
{
    <statement 1>
    .....
    <statement n>
}
```

Break and return statement can be used which results in breaking of inner loop and exiting a function respectively.

Algorithm Specification cntd..

- Pseudocode conventions:

8) A conditional statement has the following forms.

```
if <condition> then <statement>
```

```
if <condition> then <statement1> else <statement2>
```

```
Case
{
    : <condition>: <statement 1>
    .....
    : <condition>: <statement n>
    : <else>: <statement n+1>
}
```

Algorithm Specification cntd..

- Pseudocode conventions:

9) Input and Output are done using the instructions read and write.

10) There is one type of procedure : Algorithm

E.g. Algorithm Name(<parameter list>)